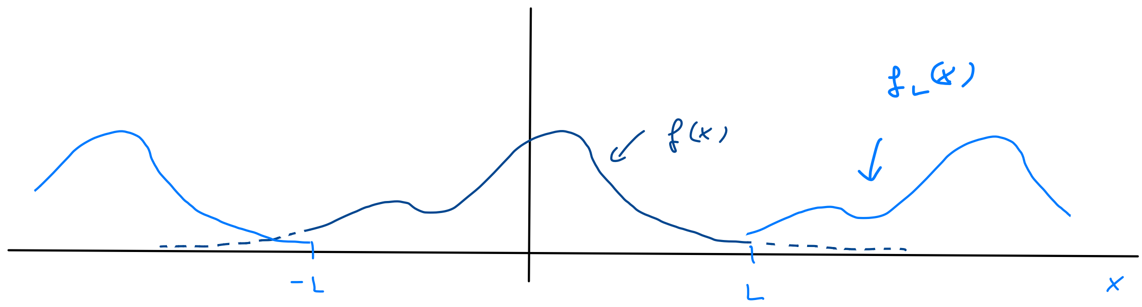
Fourier integrals

we want to extend our machinery to non-periodic functions ...

$f(x)$ nice function

① $f_L(x) = f(x)$ if $|x| < L$

② $f_L(x)$ $2L$ - periodic



use Fourier series

$$f_L(x) = a_0 + \sum_{n \geq 1}^{\infty} \left\{ a_n \cos\left(n \cdot \frac{\pi}{L} \cdot x\right) + b_n \sin\left(n \cdot \frac{\pi}{L} \cdot x\right) \right\} \qquad (1)$$

$$a_n = \frac{1}{L} \int_{-L}^{L} dy \, \cos\left(n \cdot \frac{\pi}{L} \cdot y\right) f(y) \qquad (a)$$

$$b_n = \frac{1}{L} \int_{-L}^{L} dy \, \cos\left(n \cdot \frac{\pi}{L} \cdot y\right) f(y) \qquad (b)$$

$$a_0 = \frac{1}{2L} \int_{-L}^{L} dy \, f(y) \qquad (c)$$

- rewrite cosine terms

$$\sum_{n=1}^{\infty} \underbrace{\frac{1}{L}\int_{-L}^{L} dy\, f(y)\, \cos\left(n\cdot\frac{\pi}{L}\cdot y\right)}_{a_n \text{ from (a)}} \cdot \cos\left(n\cdot\frac{\pi}{L}\cdot x\right)$$

$n\cdot\frac{\pi}{L} \equiv \omega_n$

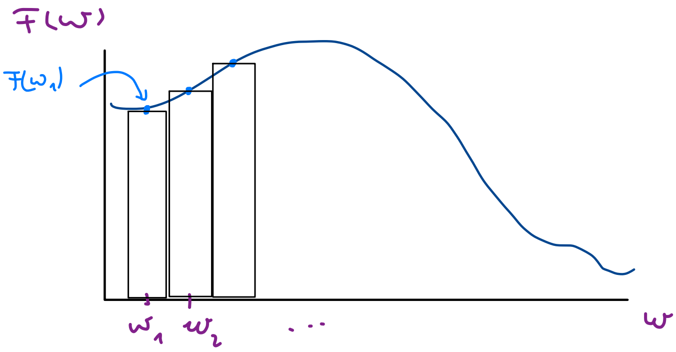
$$= \sum_{n=1}^{\infty} \frac{1}{\pi}\, \Delta\omega \int_{-L}^{L} dy\, f(y)\, \cos(\omega_n\cdot y)\cdot\cos(\omega_n\cdot x)$$

$\Delta\omega \equiv \omega_{n+1} - \omega_n = \frac{\pi}{L}$

now take $L\to\infty$:

summand $\rightarrow \int_{-\infty}^{\infty} dy\, f(y)\, \cos(\omega_n\cdot y)\cdot\cos(\omega_n\cdot x) =: F(\omega_n)$

sum of cos terms: $\Rightarrow \frac{1}{\pi}\sum_{n=1} \Delta\omega\, F(\omega_n)$ approximates integral $\frac{1}{\pi}\int_0^{\infty} d\omega\, F(\omega)$



sum of cos terms $\xrightarrow[L]{\infty}$ $\int_0^{\infty} d\omega\, \cos(\omega x)\, \frac{1}{\pi}\int_{-\infty}^{\infty} dy\, \cos(\omega\cdot y)\, f(y)$

- a_0 can be neglected

$$a_0 \rightarrow \frac{1}{2L}\int_{-L}^{L} dy\, f(y) < \frac{1}{2L}\int_{-\infty}^{\infty} dy\, |f(y)| \xrightarrow[L]{\infty} 0$$

• sum of sine's

$$\sum_{n=1}^{\infty} b_n \sin\left(n \cdot \frac{\pi}{L} \cdot x\right) \longrightarrow \frac{1}{\pi} \int_0^{\infty} dw \int_{-\infty}^{\infty} dy \, \sin(w \cdot x) \sin(w \cdot y) \cdot f(y)$$

putting all pieces together, we find

$$f_L(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(n \frac{\pi}{L} \cdot x\right) + b_n \cdot \sin\left(n \cdot \frac{\pi}{L} \cdot x\right)$$

$$\downarrow$$

$$f(x) = \frac{1}{\pi} \int_0^{\infty} dw \, \cos(w \cdot x) \int_{-\infty}^{\infty} \cos(w \cdot y) f(y) \, dy \; +$$

$$+ \frac{1}{\pi} \int_0^{\infty} dw \ldots \text{" cos } \leftrightarrow \text{ sin" } \ldots \qquad (*)$$

Sine and cosine transform

$$A(w) \equiv \frac{1}{\pi} \int_{-\infty}^{\infty} dy \, \cos(w \cdot y) \; f(y)$$ cosine transform of f

$$B(w) \equiv \frac{1}{\pi} \int_{-\infty}^{\infty} dy \, \sin(w \cdot y) \; f(y)$$ sine transform of f

these correspond to Fourier coefficients ...

$$f(y) = \int_0^{\infty} dw \left[A(w) \cos(w \cdot y) + B(w) \sin(w \cdot y)\right]$$

$$f(x) \iff A(w), \; B(w)$$

example: $$f(x) = \begin{cases} 1 & \text{if } |x| < 1 \\ 0 & |x| > 1 \end{cases}$$

$$A(\omega) = \frac{1}{\pi} \int_{-1}^{1} dy \, \cos(\omega y) \cdot 1 = \left[\frac{1}{\pi} \frac{\sin(\omega \cdot y)}{\omega} \right]_{-1}^{1} = \frac{2}{\pi} \frac{\sin(\omega)}{\omega}$$

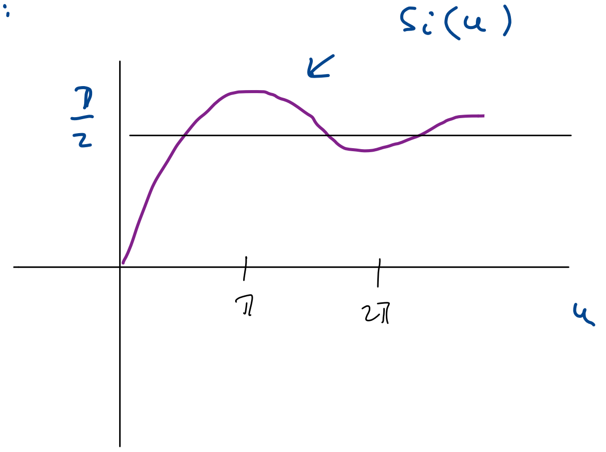
$B(\omega) = \emptyset$

$$\Rightarrow \quad f(x) = \frac{2}{\pi} \int_0^\infty \frac{\cos(\omega \cdot x) \, \sin(\omega)}{\omega} \, d\omega$$

$$x = 0 \quad \Rightarrow \quad \int_0^\infty \frac{\sin(\omega)}{\omega} \, d\omega = \frac{\pi}{2} \quad !$$

sine integral function:

$$Si(u) = \int_0^u \frac{\sin(x)}{x} \, dx$$



$Si(u) = u + \dots$ for small u

$Si(u) \xrightarrow[u]{\infty} \frac{\pi}{2}$

In general

$f(x)$ even $\Leftrightarrow$ $A(\omega)$ $\rightarrow$ cosine transform

$f(x)$ odd $\Leftrightarrow$ $B(\omega)$ $\rightarrow$ sine transform

Dirac delta

Rewrite (*)

$$f(x) = \frac{1}{\pi}\int_0^\infty dw \int_{-\infty}^{\infty} dy \, \cos(w\cdot x)\cos(w\cdot y) f(y)$$
$$+ \frac{1}{\pi}\int_0^\infty dw \int_{-\infty}^{\infty} dy \, \sin(w\cdot x)\cdot\sin(w\cdot y) f(y)$$

$$= \frac{1}{\pi}\int_0^\infty dw \int_{-\infty}^{\infty} dy \underbrace{\Big(\cos(w\cdot x)\cos(w\cdot y) + \sin(w\cdot x)\sin(w\cdot y)\Big)}_{\cos(w(x-y))} f(y)$$

$$= \frac{1}{\pi}\int_0^\infty dw \int_{-\infty}^{\infty} dy \, \cos(w(x-y))\cdot f(y)$$

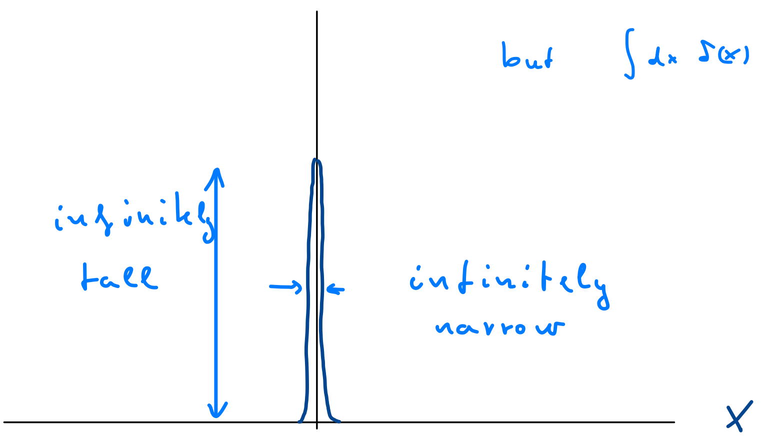
now assume, I can exchange 2 integrals as improper integrals!

$$\Rightarrow \quad f(x) = \int_{-\infty}^{\infty} dy \underbrace{\left(\int_0^\infty dw \, \cos(w(x-y))\right)}_{\substack{\parallel \\ \delta(x-y) \\ \parallel \\ \delta(y-x)}} \cdot f(y)$$

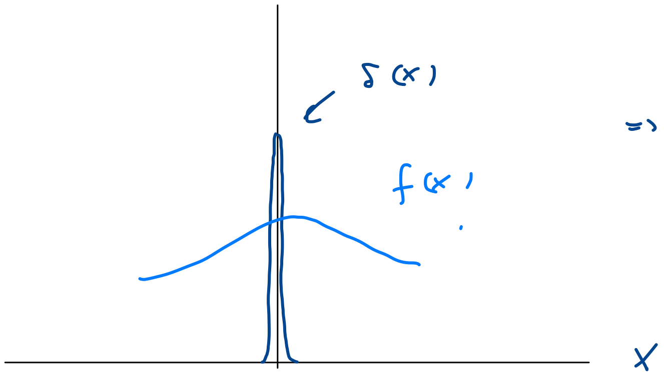
Dirac delta "function"

Dirac delta?

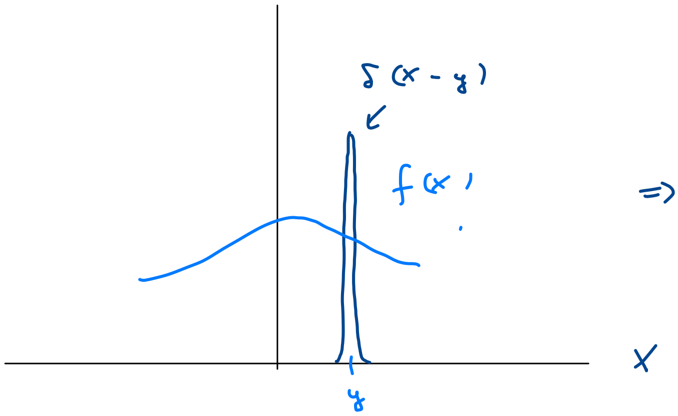
$\delta(x)$



but $\int dx \, \delta(x) = 1$!



$$\Rightarrow \int_{-\infty}^{\infty} dx\, \delta(x) \cdot f(x) = f(0)$$



$$\Rightarrow \int_{-\infty}^{\infty} dx\, \delta(x-y)\, f(x) = f(y)$$

Dirac delta is not a function, but distribution

can be represented as a limit of functions

Statement $\quad g(x)$ integrable with $\int_{-\infty}^{\infty} dx\, g(x) = 1$

$$\delta_\Omega(x) \equiv \Omega \cdot g(x \cdot \Omega)$$

- $\int_{-\infty}^{\infty} \delta_\Omega(x)\, dx = \int_{-\infty}^{\infty} dx \cdot \Omega\, g(\underbrace{x \cdot \Omega}_{y}) = \int_{-\infty}^{\infty} dy \cdot g(y) = 1$

- as $\Omega \to \infty$: $\delta_\Omega(x)$ becomes tall $\sim \Omega$ and shrinks in x direction $\sim \frac{1}{\Omega}$

- $\lim_{\Omega \to \infty} \delta_\Omega(x) = \delta(x)$!

$$\delta(x-y) \stackrel{?}{=} \frac{1}{\pi}\int_0^\infty d\omega \cdot \cos(\omega(x-y))$$

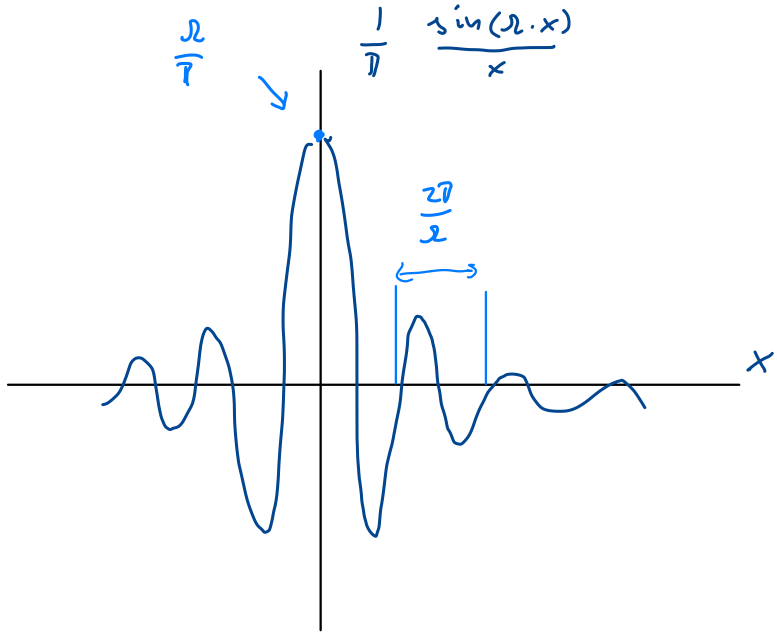
must be evaluated as improper integral

$$\delta_\Omega(x) \equiv \frac{1}{\pi}\int_0^\Omega d\omega \; \cos(\omega \cdot x) = \frac{1}{\pi}\,\frac{\sin(\Omega \cdot x)}{x}$$

in this case $\quad g(x) = \frac{1}{\pi}\frac{\sin(x)}{x}$

- $\Omega \cdot g(\Omega \cdot x) = \cancel{\Omega}\,\frac{1}{\pi}\,\frac{\sin(\Omega \cdot x)}{\cancel{\Omega} \cdot x} = \delta_\Omega(x)$ ✓
- $\int_{-\infty}^{\infty} dx\; g(x) = \frac{1}{\pi}\cdot 2 \cdot \int_0^\infty \frac{\sin(x)}{x}\,dx = \frac{2}{\pi}\cdot\frac{\pi}{2} = 1$ ✓

many representations exist for $\delta(x)$, this looks a bit ugly ...



Complex Fourier transform

$$f(x) = \frac{1}{\pi}\int_0^\infty d\omega \int_{-\infty}^{\infty} dy \; \underbrace{\cos(\omega(x-y))}_{\frac{1}{2}\left(e^{i\omega(x-y)} + e^{-i(x-y)\omega}\right)} f(y) =$$

$$= \frac{1}{2\pi}\int_0^\infty d\omega \int_{-\infty}^{\infty} dy \; e^{i\omega(x-y)} f(y) + \underbrace{\frac{1}{2\pi}\int_0^\infty d\omega \int_{-\infty}^{\infty} dy \; e^{-i\omega(x-y)} f(y)}$$

$\omega \to \tilde{\omega} \equiv -\omega$

$$\Rightarrow \frac{1}{2\pi}\int_{-\infty}^{0} d\tilde{\omega} \int_{-\infty}^{\infty} dy \; e^{i\cdot\tilde{\omega}(x-y)} f(y)$$

merge them

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} d\omega \int_{-\infty}^{\infty} dy \; \underbrace{e^{i\omega(x-y)}}_{e^{i\omega\cdot x}\cdot e^{-i\omega\cdot y}} f(y)$$

Fourier identity!

$$\boxed{f(x) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} d\omega \; e^{i\omega x} \underbrace{\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-i\omega y} f(y)\, dy}_{\hat{f}(\omega)}}$$

$\hat{f}(\omega)$ Fourier transform of $f(x)$

$$\boxed{\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} dy \; f(y)\cdot e^{-i\cdot\omega\cdot y}}$$

Fourier transformation

$$\boxed{f(y) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} d\omega \; \hat{f}(\omega)\, e^{i\omega\cdot y}}$$

inverse Fourier transformation

Does $\hat{f}(\omega)$ exist?

Statement: if $|f(x)|$ integrable and piece wise continuous on any finite interval $\Rightarrow \exists$ $\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int dx \, e^{-i\omega \cdot x} f(x)$

notation $\hat{f} = \mathcal{F}(f)$, $f = \mathcal{F}^{-1}(\hat{f})$

$f(x)$ and $\hat{f}(\omega)$ form a pair

$\mathcal{F}$ takes a function f and finds its pair $\hat{f}$

Example: $f(x) = \begin{cases} e^{-x} & x > 0 \\ 0 & \text{otherwise} \end{cases}$

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_0^\infty dx \, e^{-x} \, e^{-i\omega \cdot x} \overset{\uparrow}{=}$$

$\int e^{\alpha x} \Rightarrow \frac{1}{\alpha} e^{x\alpha}$ works for $\alpha \in \mathbb{C}$, too

$$= \frac{1}{\sqrt{2\pi}} \left[\frac{1}{-1 - i\omega} e^{-x(1+i\omega)} \right]_0^\infty = \frac{1}{\sqrt{2\pi}\,(1+i\omega)}$$

A few Fourier transforms

$f(x)$	$\mathcal{F}(f)$
1	$\sqrt{2\pi} \cdot \delta(\omega)$
$e^{-\lvert x \rvert}$	$\sqrt{\frac{2}{\pi}} \frac{1}{1+\omega^2}$
$\Theta(x)\, e^{-x}$	$\frac{1}{\sqrt{2\pi}} \frac{1}{1+i\omega}$
$e^{-\frac{x^2}{2}}$	$e^{-\omega^2/2}$
$\chi_{[-1,1]}(x)$	$\sqrt{\frac{2}{\pi}} \frac{\sin(\omega)}{\omega}$